A Preliminary Analysis of the Discriminative Capacity of the Body Roundness Index, Conventional Anthropometrics, and Diet Quality for Identifying Suicide Risk at First Consultation in Major Depression

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NATIONAL NETWORK of DEPRESSION CENTERS

BACKGROUND

- Suicide risk in major depressive disorder (MDD) is complex and still assessed subjectively.[1]
- There's controversy about whether higher body mass index (BMI) associates with lower suicidal ideation and mortality.[2]
- The Body Roundness Index (BRI) models the body as an ellipse to estimate total and visceral fat.[3]
- BRI outperforms BMI for cardiometabolic outcomes;[4] whether anthropometrics or diet quality can identify suicide risk remains unclear.

OBJECTIVE

Evaluate whether BRI, four conventional anthropometric measures and a brief diet quality screener (REAP-S) identify and grade suicide risk at the first psychiatric consultation for MDD.

METHODS

Design and sample

Cross-sectional analysis of a subset of patients with MDD at their first outpatient psychiatric consultation (n = 652).

Six predictors

BMI · WC · WHR · WHtR · BRI · REAP-S

Blue: anthropometric indices · Orange: diet quality (Rapid Eating Assessment for Participants–Shortened)

$BRI = 364.2 - 365.5 \times \sqrt{[1 - (WC/2\pi)^2/(0.5 \times height)^2]^3}$

Outcome

Suicide risk, as presence (yes/no) and severity (none/low/high), assessed by clinical ascertainment.

Analysis

1. **12 logistic models:** 6 binomial, 6 ordinal; unadjusted and adjusted for age and sex
2. **Assumptions:** Box-Tidwell, Cook's distance, Brant test
3. **Discrimination:** AUC (95% CI), alone and with age + sex
4. **Comparison:** pairwise DeLong tests, Holm-Bonferroni correction

RESULTS

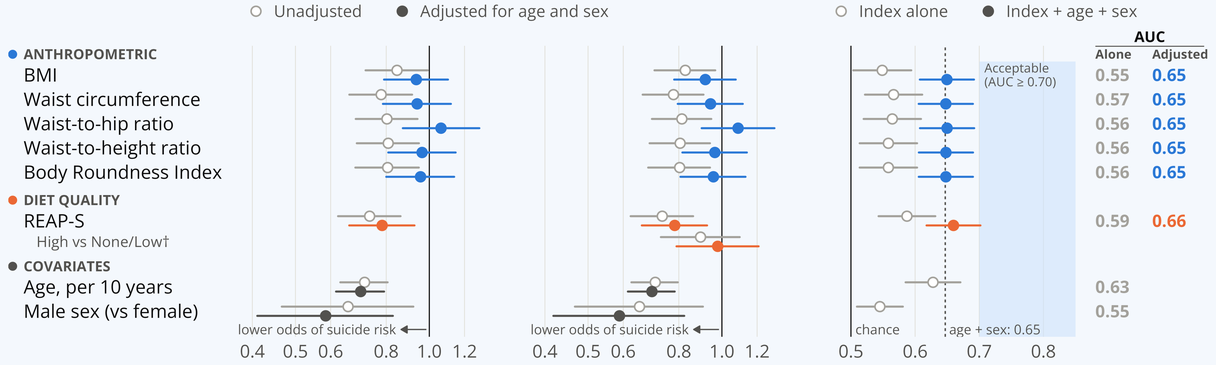
652 patients with MDD

67.8% female

25 y median age (IQR 20–39)

41.6% with suicide risk

After adjusting for age and sex, only REAP-S remains statistically significant



Holm-adjusted p < 0.05 — A: 6/6 indices unadjusted, 1/6 adjusted (REAP-S); B: 5/5 anthropometric indices unadjusted, 0/5 adjusted. Age and sex are mutually adjusted.
†REAP-S: proportional odds not met (Brant test p = 0.02), so ORs are shown per cut-point; the first cut (any risk vs none) is on the main row.
C: adding an index to age + sex changed the AUC by ≤ 0.01 (DeLong vs. age + sex, all p > 0.05).
1 SD: BMI 6.4 kg/m² · WC 15 cm · WHR 0.085 · WHtR 0.093 · BRI 2 · REAP-S 4.7 points.

Characteristic	Overall n = 652	No risk n = 381	Suicide risk n = 271	*p*
• Age, years‡	25 (20–39)	28 (21–46)	23 (20–30)	**<0.001**
• Female sex, n (%)	442 (67.8)	244 (64.0)	198 (73.1)	**0.015**
• BMI, kg/m²	27.0 (6.4)	27.4 (6.5)	26.4 (6.2)	**0.039**
• Waist circumference, cm	86.5 (15.4)	88.1 (15.8)	84.4 (14.5)	**0.002**
• Waist-to-hip ratio	0.83 (0.08)	0.84 (0.09)	0.82 (0.08)	**0.007**
• Waist-to-height ratio	0.53 (0.09)	0.54 (0.10)	0.52 (0.09)	**0.008**
• Body Roundness Index	4.1 (2.0)	4.3 (2.1)	3.9 (1.8)	**0.008**
• REAP-S score	25.1 (4.7)	25.7 (4.7)	24.3 (4.6)	**<0.001**

Mean (SD) unless stated; ‡median (IQR). Bold: p < 0.05. Welch t-test; Wilcoxon rank-sum test for age; Pearson χ^2 test for sex.

	Any suicide risk		Risk severity	
OR (95% CI)	○ Unadjusted	● Adjusted*	○ Unadjusted	● Adjusted*
• BMI	**0.85** (0.72–0.99)	0.93 (0.79–1.10)	**0.83** (0.71–0.97)	0.92 (0.78–1.07)
• Waist circumference	**0.78** (0.66–0.91)	0.94 (0.79–1.12)	**0.78** (0.66–0.91)	0.94 (0.80–1.11)
• Waist-to-hip ratio	**0.80** (0.68–0.94)	1.06 (0.87–1.30)	**0.81** (0.70–0.95)	1.09 (0.90–1.31)
• Waist-to-height ratio	**0.81** (0.69–0.95)	0.96 (0.81–1.15)	**0.81** (0.69–0.94)	0.96 (0.81–1.14)
• Body Roundness Index	**0.81** (0.68–0.95)	0.96 (0.80–1.14)	**0.80** (0.68–0.94)	0.96 (0.81–1.13)
• REAP-S	**0.73** (0.62–0.86)	**0.78** (0.66–0.93)	**0.73** (0.62–0.86)	**0.78** (0.66–0.93)
High vs None/Low†			0.90 (0.73–1.10)	0.98 (0.79–1.21)
• Age, per 10 years	**0.72** (0.63–0.81)	**0.70** (0.62–0.79)	**0.71** (0.63–0.80)	**0.70** (0.61–0.78)
• Male sex (vs female)	**0.66** (0.47–0.92)	**0.58** (0.41–0.83)	**0.65** (0.47–0.91)	**0.59** (0.42–0.82)

Indices: OR per 1-SD increase; age: per 10 years; sex: men vs women.
*Indices adjusted for age and sex; age and sex mutually adjusted. Bold: 95% CI excludes 1.
†REAP-S did not meet proportional odds (Brant test p = 0.02): severity ORs by cut-point.

TAKE-HOME MESSAGE

Body size did not signal suicide risk

Leaner patients appeared at higher risk, but the association disappeared after adjusting for age and sex. REAP-S was associated with lower odds of any suicide risk (presence / absence), but not with risk severity. No measure discriminated suicide risk (AUC ≤ 0.59).

LIMITATIONS

- Preliminary, cross-sectional, single-center analysis.
- No adjustment for depressive severity or psychotropic medication, which can affect weight, diet and suicide risk.
- REAP-S is self-reported and may partly reflect depressive symptoms such as appetite change.

CONCLUSIONS

- Neither anthropometric measures nor REAP-S discriminated suicide risk (AUC 0.55–0.59).
- Anthropometric associations with suicide risk and its severity vanished after adjusting for age and sex; anthropometric studies in psychiatry should adjust for both.
- REAP-S remained associated with the presence of suicide risk after adjusting for age and sex (aOR 0.78 per SD).
- **Next:** item-level REAP-S analysis, adjustment for depressive severity, and longitudinal follow-up.

References

1. Orsolini L, et al. Psychiatry Investig. 2020;17(3):207–21.
2. Świątecka M, et al. Qual Sport. 2026;49:67664.
3. Thomas DM, et al. Obesity. 2013;21(11):2264–71.
4. Eng PC, et al. Diabetes Obes Metab. 2025;27(10):5554–65.

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